

Thesis Proposal Draft

To be rendered in LaTeX most likely

1. Introduction and Motivation

Core points to get across:

- ☐ PLCs are typologically slippery and theoretically important; **to do**: somewhat present, needs expansion
- ☐ Existing typologies and models miss the interaction of morphology, semantics, TAM systems, and variation; **to do**: requires brief, targeted overview here, to be fleshed out in background section
- ☒ ~~The goal of this project is to build a feature-based typology with computational methods to test, refine, and scale~~

Project Overview

Perfect-like constructions (PLCs) show systematic cross-linguistic patterns shaped by morphology-semantics mapping, language-specific TAM landscapes, and potentially other interacting factors. This project aims to develop a feature-based typological model using parallel corpora and computational pipelines to evaluate these patterns across languages, combining inferential statistics (X^2 , ϕ), weighted error diagnostics, and semantic mapping (i.e. PCA) to evaluate classifier model performance. The methodology has a computational backbone (data science, NLP, machine learning), but the ultimate goal is linguistic insight: refining our understanding of PLCs, highlighting underdescribed variation, and connecting computational tools back to linguistic theory.

Aims & Research Questions

Research Aims/Goals

The specific aims/goals of this project are as follows:

Feature schema: This project will define a cross linguistic PLC feature inventory and mapping guidelines (between form and meaning), using cross-linguistic data from a typologically diverse set of languages in which the language specific forms are described? \footnote{The pilot experiment uses English, German, and Mandarin Chinese; ensuing experiments will include Georgian as a typologically interesting case study, as well as three to five additional (yet to be chosen) languages.} as represented in corpus data \footnote{The pilot experiment uses New Testament data, and ensuing experiments will incorporate TED Talks data, and model testing on novel data is planned to be performed on material from the DoReCo database.}. The data for the pilot experiment were all coded by the author (including both feature values and corresponding non-English constructions), though for subsequent experiments L1 speakers (presumably undergraduate RAs) will be recruited to ensure high quality of annotation; extensive discussion of this facet of the project is discussed in the Pilot Experiment section.

Cross-disciplinary integration: This project is explicitly positioned such as to bridge the fields of (formal) linguistics and computational/data science, ultimately demonstrating how computational pipelines can sharpen linguistic questions, while linguistic theory can enrich computational models. Essentially, this project aims to articulate a roadmap for collaboration between fields, especially around TAM and cross-linguistic generalization.

Ontological innovation: This project is conceived of as a pilot study in applying linguistic ontologies to machine learning/NLP. By testing a feature-based ontology on PLCs — a notoriously complex semantic domain — the project demonstrates how ontological feature schemas can mediate between linguistic theory and computational modeling. This opens pathways for future research applying similar methods to other phenomena that have thus far proven challenging for existing LLMs and NLP systems, such as distinguishing embedded versus nominal clauses, robustly modeling linguistic nuance in areas like TAM and linguistic ambiguity, and perhaps most importantly improving accessibility of digital resources to low-resource, underdescribed, and/or typologically diverse languages. Ontology-driven feature schemas and modeling pipelines tested on PLCs could serve as a prototype for tackling these problems and contributing to a broader program whereby linguistic ontologies facilitate cross-disciplinary advances.

Key/Core Research Questions

Core points to get across:

- ☐ **Core question:** To what extent can a feature-based semantic ontology, developed from aligned parallel corpus data, model cross-linguistic variation in PLCs?
- ☐ **Core question:** Does the application of such a schema reveal typologically consistent contrasts in meaning (e.g. experientiality, current relevance, etc.) across languages that differ morphosyntactically and genealogically, offering a scalable and interpretable framework for typological comparison, semantic annotation, and the representation of temporally complex meanings in both research and applied settings?
- ☐ **Concomitant questions:**
 - ☐ Is the feature-based schema internally coherent and usable?
 - ☐ Can these features be defined clearly?
 - ☐ Are they recognizable across languages? (or statistically or typologically aligned subgroups of languages)
 - ☐ Can they be reliably annotated?
 - ☐ How do individual languages express these features?
 - ☐ What features do English PLCs encode versus other languages' PLCs?
 - ☐ Are there recurring patterns? Shared strategies?
 - ☐ Can we group languages or constructions by their feature “profiles”?

Background & Lit Review

Core points to get across:

- ☐ Formal / semantic approaches (Klein, Iatridou, Pancheva, etc., interface debates [Aktionsart, TAM, universals vs existentials, etc.]).
- ☐ Typological surveys (Dahl, Plungian, etc.).
- ☐ Computational / corpus approaches (Chen, van der Klis, etc.).
- ☐ Gap: no integrated, ontology-driven method that bridges linguistics + computation.

Formal/Semantic Approaches

Definitely discuss:

- ☐ Alexiadou, Rathert, & von Stechow (2003)
- ☐ Anagnostopoulou (1998)
- ☐ Grønn & von Stechow (2021)
- ☐ Iatridou et al. (2001)
- ☐ Klein (1992)
- ☐ Löbner (2002, “Is the German Perfekt a perfect perfect?”)
- ☐ McCawley (1981)
- ☐ McCoard (1976)
- ☐ Mittwoch (2008)
- ☐ Nishiyama Keonig (2010)
- ☐ Pancheva (2003)

- ☐ Pancheva (2004)
- ☐ Pancheva & von Stechow (2004)
- ☐ Portner (2003)
- ☐ Portner (2011)
- ☐ Ritz (2012)
- ☐ Rothstein (2006)
- ☐ Schaden (2009)
- ☐ Stowell (2007)
- ☐ Stowell (2008)
- ☐ Zhao (2022a, 2022b)

Maybe discuss:

- ☐ Abusch & Rooth (1990)
- ☐ Carrasco (2015 “Perfect states”)
- ☐ Declerck (2006, “The absolute use of the present perfect”)
- ☐ Franco & Tagliamonte (2022)
- ☐ Iatridou (2017)
- ☐ Izvorski (1997, “The present perfect as an epistemic modal”)
- ☐ Kamp et al. (2015)
- ☐ Katz (2011)
- ☐ Kiparsky (2002)
- ☐ Kratzer (2004)
- ☐ Lindstedt (2000)
- ☐ Matthewson et al. (“Inchoativity...”)
- ☐ Paslawaska & von Stechow (2003, “Perfect readings in Russian”)
- ☐ Werner (2013, “Temporal adverbials and the present perfect/past tense alternation”)

Typology and Typological Surveys

- ☐ Bertrand et al. (2022)
- ☐ Bowler & Ozkan (2017)
- ☐ Dahl (2000, the PFQ)
- ☐ Dahl & Velupillai (2013)
- ☐ de Swart (2016)
- ☐ Haspelmath:
 - ☐ 2010a: “Comp concepts and desc cats in crossling studies”
 - ☐ 2010b: “The interplay btwn comp concepts and descr cats”
 - ☐ 2016: “The challenge of making lang desc and comp mutually beneficial”
 - ☐ 2021: “Explaining gram coding asym form-freq corr and predictability”
 - ☐ 2023: “Word-class universals and lang-particular analysis”
- ☐ Matthewson et al. (2014, “Miss Smith’s Bad Day”)
- ☐ Plungian (2011)

Computational/Corpus Approaches

- ☐ Chen et al. (2021)
- ☐ LeBruyn, van der Klis, & de Swart (2019, “The perfect in dialog”)
- ☐ van der Klis et al. (2017)
- ☐ van der Klis et al. (2021)
- ☐ van der Klis, LeBruyn, & de Swart (2021, “The present perfect: an unstable category in an overall stable tense-aspect grammar of the western European languages”)

Gap

Where does this project fit in? What issues does it seek to resolve? What are its contributions to the literature?

IV. Pilot Experiment

This section presents the pilot experiment which has been completed, and which informs and shapes the following work of the project.

Methodology

The data used in the pilot experiment are comprised of 300 tokens in three languages; the tokens are all pulled from New Testament Bibles (NTBs), and the languages in the experiment are English, German, and Chinese. Each token represents a distinct instance of a construction of interest (i.e. each unique occurrence of a PLC); these tokens were identified in a tokenized NTB using a simple Python script which locates, extracts, and compiles all instances of auxiliary *have* in co-occurrence with a verbal noun (as defined by spaCy; these instances are almost exclusively past passive participles) in the English NTB. The output of this script is a collection of verses, indexed by verse number, which contain one or more examples of a PLC. Next, the verse numbers are aligned with identically indexed verses in NTBs in the additional language of the study¹.

Once the data are extracted and aligned, the next step is annotation; note that here *annotation* refers to the process of examining each individual PLC in its contexts and coding it for semantic/ontological features and identifying which construction is used in each language. The primary disadvantage of this process, at least at this stage, is that as an L1 speaker of only one language (English), the experimenter is limited in their ability to identify semantic nuance in the non-English constructions, such that true equivalence (whether the aligned text encodes perfect-like semantics) and any additional semantic features (such as grammatically or pragmatically encoded evidentiality) are left to the annotator’s best interpretations. In subsequent studies, this will be addressed by recruiting undergraduate RA native speakers to conduct the non-English coded work; these RAs will work in tandem with the author using a clear and concise annotation manual and performing checks via inter-annotator agreement (IAA).

¹ The obvious limitation of this methodology is that it only identifies PLCs that encoded as *have*+participle perfects in English; subsequent experiments will use reverse lookup techniques to identify constructions of interest in each of the other languages as well.

Features

The features used in the pilot experiment are all drawn directly from the formal semantic literature on perfects, and represent some of the most commonly cited meaning components associated with PLCs cross-linguistically. Note that one of the most commonly cited of these meaning components is what is often referred to as “current relevance”, or sometimes “discourse salience”, i.e. the implication that the event or state encoded by the predicate which is framed as a PLC carries some specific relevance for the present discourse. This feature is generally associated specifically with present perfects, and is arguably a highly salient component of their meaning in some languages (including English); however, as discussed by Grønn and von Stechow (2021), there has as yet been no widely agreed upon formal semantic definition of this feature, and there is open debate among researchers as to whether current relevance is even part of the truth conditional semantics of the perfect. For these reasons, this feature was not included in the pilot experiment, though its incorporation in subsequent experiments — in some form or other — remains a goal.

Each feature used in the pilot experiment is presented here along with the operationalized definition used to evaluate its presence or absence in each individual token. Further discussion of the performance and impact of performance on feature scheme refinement is discussed in depth in the Results and Discussion section below.

Experiential Reading

A construction expresses an experiential reading if it conveys that at least one instance of the event type denoted by the predicate has occurred within the relevant time frame, typically the speaker’s past experience. The event must be completed prior to the reference time (assumed to be the speech time), and its occurrence may be inferred or explicitly stated.

Resultative Reading

A construction expresses a resultative reading if it encodes or implies a state that results from a change-of-state event, and that state holds up to (and potentially includes) the reference time (typically speech time).

Universal Reading

Operationalized definition: A construction expresses a universal reading if it conveys that a state or event holds continuously throughout a contextually defined time span. This reading often implies present-tense relevance but need not extend to the moment of utterance.

Compatibility with DTAM²

A construction is compatible with DTAM if it occurs (without marked infelicity) alongside a past-referencing definite temporal adverbial (e.g., yesterday, in 1999, last week). This is judged based on corpus occurrence or contextual plausibility.

Due to the nature of the data extraction process used in the pilot experiment, all present perfects in this data set are negative for this feature, as the English present perfect (in most dialects) is not compatible with DTAM. This will be addressed in subsequent experiments through the use of reverse lookup, as discussed in the introduction to this section, above.

² DTAM = Definite (Past-Referencing) Temporal Adverbial Modification, the crux of the well-known “present perfect puzzle” (Klein 1992)

Evidentiality

A construction expresses evidentiality if it overtly marks the source of information, whether through grammatical markers (*reportedly, apparently*) or lexical/discourse framing (*I heard, they say*). Note that evidentiality does not require topical continuity but rather signals the epistemic stance of the speaker toward the event.

As evidentiality is not grammatically encoded in the English TAM system, the handful of instances of this feature in the pilot experimental dataset all either explicitly encode source of knowledge (such as “he saw that [predicate]”) or otherwise make source of knowledge explicit in the discourse context. Several of the languages that are marked for possible inclusion in subsequent experiments do mark evidentiality, or rather are widely noted to include evidentiality in their truth conditional semantics (e.g. Georgian³, Turkish⁴, and Bulgarian⁵), thus it is deemed worthwhile to maintain its inclusion in the feature set.

TAM Layering

A construction incorporates TAM layering if it combines two or more tense-aspect-modality components (e.g., perfect + past, perfect + progressive, perfect + modal). In the pilot experiment this feature was a major source of classifier prediction errors; this will be discussed in more depth in the Results and Discussion section below.

Negation/Polarity Effects

A token is marked as expressing negation if the construction appears under the scope of negation — lexically, morphologically, or syntactically. This is a surface-level annotation used to examine whether construction choice varies with polarity.

“Unofficial” Features

In addition to the features listed here which were all coded in the data fed to the classifier experiments, a variety of additional features of potential interest were specifically catalogued in the metadata should they prove to be relevant to subsequent experiments; these include: interrogatives, counterfactuals, causatives, passives, and inchoatives. At this remove it is not yet clear to what extent these “contextual” potential features warrant inclusion in the feature set; this remains an open question, and input and suggestions are very welcome.

Construction Identification

One of the greatest — but also most intriguing, at least for this morphology-obsessed researcher — challenges in the coding process was identifying which construction was used in each language, how to group (or split) those constructions into categories, as well as whether an aligned token in the non-English data was “equivalent”, meaning that it encoded the core (especially temporal) semantics that are associated with PLCs⁶.

³ Widely cited, incl. Makharoblidze (2012), Kurtsikidze (2006), Bolkvadze & Kiziria (2023)

⁴ Şener (2011), de Gül (2008), Rett (2013)

⁵ Rett & Hyams (2013)

⁶ I recognize that this framing is vague; what are the “core” semantics of PLCs? Is that not one of the goals of this project? For now, I use the informal diagnostic of encoding some degree of at least partial anteriority, especially when further reinforced by a distinction between event/initialization time and reference time. This remains one of the major challenges in strategizing subsequent experiments, and input and suggestions are very welcome.

Running the Experiments

Once all of the data had been coded, they were fed into two classifier models for experimentation: Random Forests (“RFC”) and MLPClassifier (“MLP”). Random Forests operates on a decision tree model, while MLPClassifier (Multi-Layer Perceptron) is a neural network; using the two classifiers in tandem allows for examining the performance of the feature set using two distinct model types, thus providing greater context for evaluation of their salience and framing.

Both experiments use the same strategy for loading in the data: the 300 tokens were split into a 150-token training set and a 150-token testing set. Although this pilot split was random rather than stratified, subsequent experiments will explicitly stratify by feature to ensure proportional distribution. The classifiers then used the 150-token training set to “learn” the features and correlation patterns with occurrence of constructions, and based on that training attempted to predict which constructions would be used in test set tokens. The results of the predictions were then exported for use in error analysis. Note that both the German and Chinese data sets included tokens coded as “not applicable”, indicating that the token for that language was not usable for some reason; in subsequent experiments this binary applicable/ not-applicable distinction will be further decomposed into at least two subtypes of N/A: tokens in which a given verb/predicate is simply absent in that language’s text (“true” N/A), and tokens in which it is unclear whether the corresponding token “truly” encodes a PLC.

Error Analysis

Error analysis consisted of an array of data tabulation, visualization, and evaluation strategies; these are presented in the following subsections. Note that several additional methodologies were applied (e.g. error-to-feature ratio heat maps), but the results presented here were all produced using the methodologies specifically described in this section.

Classification Reports

Each language-classifier pair produced a classification report which included precision, recall, f-1 score, and support results for each construction used in the data set, as well as accuracy, macro average, and weighted average scores for each of these categories. These reports are useful for evaluating a given classifier’s overall performance on classifying each construction in each language. The **precision** value is a fraction which represents the ratio of relevant retrieved instances (of a given construction) to total retrieved instances. The **recall** value is a fraction which represents the ratio of relevant retrieved instances to total relevant instances. The **f-1 score** is the harmonic mean of the precision and recall values. **Support** refers to the total number of instances of a given construction in the test data set.

Confusion Matrices

Each language-classifier pair also produced a confusion matrix, which tabulates in matrix form the relationship between actual constructions (rows) and predicted constructions (columns). The result is a matrix representation of results, both accurate and erroneous, for each language-classification pair. This matrix allows for visualizing the ratio of erroneous to accurate predictions made by the model; to these ratios a weighted score is calculated to account for frequency effects, such that, for example construction 1 may have a 100% error rate based on 1 representative token, whereas construction 2 may have a 50% error rate based on 75 representative tokens, suggesting that the construction 2 error rate is more informative than that for construction 1.

Error-Feature Reports

For each language-classifier pair, a report was constructed to tabulate the feature values of each specific error. In this report, each row represents an individual error, displaying the token number of the error, the predicted construction, and the actual construction; each column then shows the feature values assigned to each construction. From here, the error types can be sorted and grouped, providing a simple visualization of possible correlations between feature sets and error types. For example, in the RFC for Chinese, it was shown that all errors of predicted *le* → actual *suo...de* were resultatives and predominantly accomplishments, while all errors of predicted *le* → actual unmarked were resultatives and predominantly achievements.

Putting it All Together

These data were then tabulated and submitted to various statistical analyses, the results of which are presented in the following section, Results and Discussion. These tests included: χ^2 values, χ^2 patterns by way of p -values, ϕ -values, and weighted error rate scores. Using these values in tandem allows for error analysis of considerable richness, taking into account such useful metrics as error-to-feature ratio patterns, significance of features in error patterns, and strength of association between feature paradigms and error types. The results are present below.

Results and Discussion

Overview

Before diving into language-specific results, one generalization can be made regarding both the models' performance and also the methodology that produced the model inputs. The overall error rates by language were, perhaps unsurprisingly, quite different. The English data sets both contained all 150 tokens, and yielded overall error rates of 11/150 or 7.3% (RFC) and 13/150 or 8.7% (MLP), indicating relatively strong performance. German performed rather less well, with error rates of 36/149 or 24.2% (RFC) and 35/148 or 23.6% (MLP). Both models performed significantly worse on the Chinese data, yielding 52/115 or 45.2% (RFC) and 60/121 or 49.6% (MLP). There are a variety of reasons why this might be the case, but here I will highlight two specifically: anchor language effects and feature landscape effects.

Anchor Language Effects

Because English was used as the anchor language for this experiment, naturally every English token in the initial 300-token set was included, unlike with the German and Chinese which in some cases were deemed “non-equivalent”. Even more significant, however, is the fact that the feature values for each token were assigned based on the English anchor text, meaning that the mapping between feature paradigm and construction in English is generally reliable; certainly there are ambiguous and other edge cases, but by and large the diagnostics used to evaluate feature values for each token are fairly well fleshed out and presumed to more or less accurately represent the features encoded by each token's PLC.

By contrast, the German and to an even greater extent the Chinese data presumably encode feature nuances that are not present in the English data, and perhaps even unrepresented in the feature set itself. Because German is relatively close to English both genetically and typologically, feature recovery was easier, which likely explains its intermediate performance.

Chinese, on the other hand, is wholly unrelated to English (or German), and its TAM system operates on a logic which is entirely distinct from what is seen in Indo-European languages generally, to say nothing of Low Germanic

languages specifically. Thus it is unsurprising that not only is the token count significantly lower for Chinese, at 115 and 121, but the models' performance on these data is considerably worse, nearing 50%.

Subsequent experiments will seek to counteract anchor language effects through a combination of reserve lookup extraction (i.e. searching for potential PLCs in each language, not just English), recruitment of L1 speaker annotators, and refinement of the feature set.

Feature Landscape Effects

As mentioned in the previous section, with respect to the Chinese TAM system, it generally goes without saying that each language's TAM (in fact, entire grammatical system) operates on its own logic, including what features it chooses to contrast or mark overtly versus those it doesn't, the number of TAM constructions in the system's inventory, and more. Because the bulk of the research conducted on PLCs to date has focused on Indo-European, and especially Germanic languages (English alone represents a significant focus of PLC research historically), the feature set that was developed for the pilot experiment, reflecting research landscape, tends to better reflect the types of features that those languages "care about". For example, if the English perfect is considered a "complex tense"⁷, i.e. in the Reichenbachian (1947) sense of a grammatical form which combines tense and aspect, or which divides its event and reference times, to what extent can we truly apply the logic developed for English to a language which, like Chinese, has been called "tenseless" (Lin 2006 and 2010, Collart 2021)?

Tackling this quandary is one of the central aims of this project, by way of developing a feature set which can be used effectively on cross-linguistic and typologically varied languages.

In the next section I present the results by language along with some discussion, followed by some global/cross-linguistic performance discussion.

Results and Interpretations by Language

Below I present the performance of the models by language broken into two categories: error types and feature-value associations. The error type category describes the occurrence patterns by specific error type, such as "predicted *le* → actual *unmarked*". The feature value associations incorporate a variety of statistical methodologies to better inform analysis of error patterns by incorporating feature values. In this section I present the relevant statistic information including feature error rates, *p*-values, ϕ -values, and weighted scores (informing overall feature-based performance). In each language section the quantitative results are presented first (by category), followed by interpretations and discussion of those results. A subsequent section will take all of these data and interpretation collectively to pose global interpretations and discussion.

With respect to the X^2 analyses, the following calculations were used to generate the values presented here:

- X^2 *p*-values: the *p*-values were calculated along a 2x2 axis comparing positive and negative occurrence of features and errors; i.e. *f+* *e+* represents the number of tokens that bear a feature and were predicted erroneously, while *f-* *e+* represents the number of tokens that do not bear a feature and were predicted erroneously. This analysis incorporated the Yates correction, which is designed to correct for smaller sample sizes to avoid overstating significance values.
- ϕ -values: the ϕ -values were calculated using the square root of the value of the X^2 -score divided by the total sample size for that table.

⁷ This is my own term, but I'm quite fond of it. I am unsure to what extent it is used elsewhere in the literature; I have seen it, but not a lot.

- Weighted scores (WS): these encompass a range of calculations:
 - Baseline error rate (BER): total errors / total tokens (the “expected” error rate if errors were evenly distributed across features)
 - Feature-specific error rate (FER): errors that bear the feature / total occurrences of the feature in the data set
 - Feature prevalence (PREV): total occurrences of the feature in the data set / total tokens
 - Base weighted score (BWS): $(FER - BER) * PREV$; the subtraction compares “how hard this feature is” relative to the model’s average performance, and the multiplication by prevalence scales that difference by how frequent the feature actually is in the dataset
 - Weighted score rescaled (WSR): $WS * 100$; this scales the score to be readily interpretable
 - Range-normalized score (RNS): $(WSR - \text{MIN}(WSR \text{ for that language-model pair})) / (\text{MAX}(WSR \text{ for that language-model pair}) - \text{MIN}(WSR \text{ for that language-model pair}))$; this places a weighted score in the context of the range of weighted scores for that language-model pair⁸, yielding a 0-1 value

With respect to interpretation of these values,

- The p -value indicates the statistical reliability of an association between feature(s) and error(s).
- The ϕ -value indicates the strength of association between feature(s) and error(s), independent of sample size; the scale used presumes 0-0.29 = weak association, 0.3-0.49 = medium association, 0.5+ = strong association.
- The weighted score (WS) and range of WS’s represent relative difficulty (for the models’ predictive efforts); this value can be positive or negative, wherein a positive value indicates the model struggled *harder* than baseline on a feature, while a negative score indicates that the model did *better* than baseline on a feature. In other words, the WS measures impact relative to baseline, scaled by how frequent the feature is, and is sensitive to both error bias and how frequently the feature occurs (i.e. the practical importance of a feature in the data set).
- The range-normalized score (RNS) makes results cross-comparable across languages/models with very different numeric ranges; the higher a value the worse a model performed and vice versa.
- Taking the ϕ -value and WS together allows us to evaluate both how strongly a feature is associated with predictive errors (ϕ) and the degree to which it matters in practice (WS).

Chinese

The overall error rates for the two Chinese experiments were: 45.2% of 115 tokens for RFC and 49.6% of 121 tokens for MLP.

Error Type Results

With respect to specific error types, three noteworthy patterns emerged in the results from both classifiers: predicted *le* → actual *suo...de*, predicted *le* → actual *unmarked*, and predicted *le* → actual [adverb] *le*. A further pattern emerged specifically in the RFC experiment: *unmarked* → [varies, all involving *you*], and in the MLP experiment: *suo...de* → [varies].

***Le* → *unmarked*.** This error represents 15.4% of all RFC errors, 25.8% of all *le* overpredictions, and 80% of all *unmarked* underpredictions. Similarly, MLP’s performance showed this error type representing 13.3% of all errors, 24.2% of all *le* overpredictions, and 72.7% of all *unmarked* underpredictions. In both models, each of these errors

⁸ For example, a Chinese RFC resultative score of -5.68 is quite good within the -5.68 to 4.31 range of Chinese RFC feature scores, while the German RFC experiential score of -5.67 is rather poor within the -22.90 to -5.67 range of German RFC feature scores.

was a resultative, and favored achievements at a rate of 6/8 for both models, the remainder being accomplishments.

Le → **suo...de**. This error represents 13.5% of all RFC errors, 22.6% of all *le* overpredictions, and 100% of all *suo...de* underpredictions, and in fact 100% of all instances of actual *suo...de* were *le* predictions. Similarly, MLP's performance showed this error type representing 13.3% percent of all errors, 24.2% of all *le* overpredictions, as well as 88.9% of all *suo...de* underpredictions (MLP correctly predicted *suo...de* once out of nine occurrences in the data set). In both models, each of these errors was a resultative, and favored accomplishments at a rate 5/7 for RFC and 6/8 for MLP, in both cases the remainder were achievements.

Le → **[adverb] le**. This error occurred at a similar rate as both the previous error types for both models; however, based on the nature of the error type, it can be argued that this is not a "true" error, as all "mispredictions" did in fact involve actual *le*, only with the addition of a temporal adverbial. For both models, these errors were primarily resultatives with a singular exception being experiential, though as this neatly reflects the distribution of these two features in the data, the relevance of this feature is somewhat lessened. Further, the errors were evenly split between accomplishments and achievements, and all other feature values were 0.

Unmarked → **[varies, incl. you]**. This error represents 11.5% of all RFC errors. In each of these errors the model predicted *unmarked*, and while the actuals varied, each involved the particle *yǒu* (roughly 'have'). The features are mostly generally random in their distribution, though notably there were no resultatives in the set: the tokens were evenly split between experientials and universals.

Suo...de → **[varies]**. This error represents 23.3% of all MLP errors. Of the varying actuals, three were *guo*, five were *le*, and the remainder included *unmarked*, *meiyou...guo*, and *zaoyi*. Most notably about these errors is that each token in this group is an experiential, though the remaining feature values appear random.

The remaining errors in each experiment, comprising 44.2% of RFC errors and 35% of MLP errors, displayed no discernible patterns of significance.

Feature-Value Results

A feature-value analysis on the results of both models revealed two features of potential significance: resultative (Res) and experiential (Exp).

RFC

The WS range for the Chinese RFC experiment was -5.68 to 4.31 (9.99), indicating that the model's errors cluster strongly on a couple of features (Exp = very hard, Res = very easy), while the rest are close to baseline. This suggests that Chinese error patterns are driven by a small number of critical feature contrasts, rather than widespread sensitivity.

Res feature results:

- FER: 37.3%
- WS: -5.68
- RNS: 0.00
- *p*-value: 0.011695
- ϕ -value: 0.2546

Exp feature results:

- FER: 70.0%
- WS: 4.31
- RSN: 1.00
- *p*-value: 0.027602
- ϕ -value: 0.2285

These results indicate that RFC performed best on resultatives and weakest on experientials, in fact Res with a WS of 0.00 was the strongest performing feature for this experiment while Exp with a WS of 1.00 was the weakest. Considered in tandem with the p - and ϕ -values we see that Exp is more salient to the model than Res, though the strength of association in both cases is relatively weak ($\phi < 0.3$).

MLP

The WS range for the Chinese MLP experiment was -6.73 to 5.46 (12.20), indicating that as compared to RFC, there are more features which show meaningful deviation from baseline. This suggests that MLP is more feature-sensitive in Chinese than RFC, distributing its difficulty more broadly.

Res feature results:

- FER: 40.0%
- WS: -6.73
- RNS: 0.00
- p -value: 0.00235
- ϕ -value: 0.2946

Exp feature results:

- FER: 74.1%
- WS: 5.46
- RSN: 1.00
- p -value: 0.007608
- ϕ -value: 0.2625

Rough interpretation. These results, similarly to the RFC results, suggest that RFC performed bests on resultatives and weakest on experientials. Considered in tandem with the p - and ϕ -values we see that Res and Exp are nearly equally salient to the model, and while the strength of association for Exp is still weak, for Res it nearly breaches the threshold of medium strength..

Interpretation & Discussion

With respect to error types, as noted above, the $le \rightarrow$ [adverb] le in fact are not “true” errors; the models are both correctly predicting the occurrence of the particle le , while the relative infrequency of individual [adverb] le tokens, each of which was coded uniquely in the data set, no doubt correlates with the underprediction of their occurrence. One possible interpretation is that adverbs do not contribute significantly to semantic feature expression/interpretation, though another interpretation might be that adverbs can be lumped together, though a proper approach to this would include looking at all of the adverb cases in the 150-token set for a more robust picture to inform *how* they can be lumped together.

The fact that $le \rightarrow suo...de$ are predominantly accomplishments and $le \rightarrow unmarked$ are predominantly achievements is noteworthy, though considering the tabulated featural distribution results (including statistical measures) suggest that Aktionsart is not a significant feature in the Chinese data, so for the time being I remain judicious with claims regarding their significance for the models.

The $unmarked \rightarrow$ [varies, all incl. *you*] error is noteworthy for appearing only in the RFC experiment data, and also for lacking tokens bearing the Res feature (they are evenly split between experientials and universals), a feature which is born by 83 of 115 (72.2%) tokens in the data set. This may suggest that the Chinese TAM system makes use of a +Res/-Res contrast in which *you* is never associated with +Res, though more data and testing are required to reinforce this hypothesis.

The $suo...de \rightarrow$ [varies] error is likewise noteworthy for appearing only in the MLP experiment data, and also for the fact that every token in the set is an experiential such that it represents 55.9% of all Exp prediction errors (i.e. errors that bear the Exp feature). Taken in tandem with a quite low p -value for the Exp feature in the Chinese MLP

data set, this all suggests that experientiality is highly salient to this model, though any specific claims must remain judicious in light of this feature's relatively low ϕ -value.

Similarly, while the *le* $\rightarrow$ *unmarked* and *le* $\rightarrow$ *suo...de* errors suggest a correlation with Aktionsart (accomplishment and achievement specifically), the Aktionsart *p*-values for both models did not indicate feature significance. It is possible that further testing involving a larger data set and a refined feature schema may illuminate the role of Aktionsart in the Chinese TAM system.

While not presented in depth here, the adjoined data tables show that the *unmarked* $\rightarrow$ *le* (RFC) and *unmarked* $\rightarrow$ (varies) (MLP) all exhibit relatively random feature distribution. It is potentially interesting in and of itself that the two models encountered different error patterns here; this observation can inform how the feature set is developed, though exactly *how* is still an open question⁹.

Speaking generally to both models' performance on resultatives (better than baseline) and experientials (worse than baseline), it bears noting that these features are in nearly complementary distribution in the data set, the exceptions being the 15/300 tokens that are ambiguous Exp/Res or 3/115 in the Chinese experiment specifically. This may indicate that rather than distinguishing between resultatives and experiences, the models are distinguishing between resultatives, which they handle well, and everything else, a group within which they are unable to derive nuanced distinctions based on the current feature set.

From a purely linguistic perspective, the results of both experiments suggest that the current feature schema is inadequate to the task of mapping the Chinese PLC landscape. One likely reason for this is that, as noted above, the majority of the research which has been conducted on PLCs has focused on Indo-European, and especially Germanic languages. As such, this is an area in which a project like this might be able to expand our understanding of how typologically diverse languages delineate contrasts and determine which underlying semantic features are most relevant (as reflected in the grammatical marking system) to a given TAM system. Subsequent experiments will incorporate this linguistic facet into the design logic of the feature schema, as informed by what language-specific TAM literature is available¹⁰.

German

The overall error rates for the two German experiments were: 24.2% of 149 tokens for RFC and 23.6% of 148 tokens for MLP.

Error Type Results

With respect to specific error types, three noteworthy patterns emerged in the results from both classifiers: predicted *pastperf* $\rightarrow$ actual *k2_pastperf*, predicted *pastperf* $\rightarrow$ actual *past*. and predicted *perf* $\rightarrow$ actual [varies]. In addition to these, several minor patterns appeared in the form of subgroup within the *perf* $\rightarrow$ [varies] specifically of the type *perf* $\rightarrow$ *pres*, a small cluster of predicted *pres* $\rightarrow$ actual [varies], and specifically in the MLP experiment predicted *pres* $\rightarrow$ actual *perf*.

***Pastperf* $\rightarrow$ *k2_pastperf*.** (Note that here “pastperf” refers to what in German is called the “Plusquamperfekt”, and “k2_pastperf” in German is the “Konjunktiv II Plusquamperfekt”.) This error type represents 14.3% of all RFC

⁹ This of course is one of the core questions of the thesis: what shape should a feature schema take in order to best inform model performance, and what does such a feature schema tell us about the nature of these constructions cross-linguistically.

¹⁰ Note that this observation was absolutely included in the design logic of the feature schema used in the pilot experiments, though in the interest of “casting a wide net” the feature set was kept relatively simple.

errors and 17.1% of all MLP errors. Every token of this error type bears the Res and TAML features, and both favor achievements at a rate of 5/7 for RFC and 5/6 for MLP. The remaining features were more or less evenly distributed.

Pastperf → **past**. (Note that here “past” refers to what in German is called the “Präteritum”.) This error type represents 22.2% of all RFC errors and 22.9% of all MLP errors. As with the *pastperf* → *k2_pastperf* error type, each of these tokens bears the Res and TAML features, though are evenly split between accomplishments and achievements.

Perf → **[varies]**. This error type represents 30.6% of all RFC errors and 31.4% of all MLP errors. In both experiments, the values for all features appear randomly distributed.

Perf → **pres subgroup**. While not a major error type of its own, comprising only 11.1% of all RFC errors and 11.4% of all MLP errors, this subgroup is notable in that all tokens of this error type are universals and activities.

Pres → **[varies]**. Comprising only 11.1% of all RFC and 31.4% of all MLP errors, this error type is notable in that every token in the set is a universal and a state.

Pres → **perf**. Occurring specifically with the MLP experiment, where it 11.4% of all MLP errors, this error type is notable, especially considered alongside the previous two error types, in that every token is a universal and a state.

The remaining errors in each experiment, comprising 16.7% of RFC errors and 16.7% of MLP errors, displayed no discernible patterns of significance.

Feature-Value Results

A feature-value analysis on the results of both models revealed six features of potential significance: universal (Univ), TAML, negative polarity (NegPol), activity, accomplishment (accomp), and state.

RFC

The WS range for the German RFC experiment was -22.90 to -0.00 (22.90), indicating that all features are below (better than) baseline, but the distance between the easiest (Res) and the least (Evid) is vast. These results suggest a very uneven degree of ease: some features (Res) are disproportionately easy, others only slightly easier (than baseline), pointing to a strong feature-level asymmetry.

Univ results:

- FER: 73.3%
- WS: -3.15
- RNS: 0.86
- p -value: 0.00001
- ϕ -value: 0.3943

Activity results:

- FER: 75.0%
- WS: -1.47
- RNS: 0.90
- p -value: 0.001908
- ϕ -value: 0.2894

TAML results:

- FER: 63.6%
- WS: -6.72
- RNS: 0.71
- p -value: 0.00001
- ϕ -value: 0.5051

Accomp results:

- FER: 12.6%
- WS: -13.03
- RNS: 0.43
- p -value: 0.017441
- ϕ -value: 0.2108

NegPol results:

- FER: 54.5%
- WS: -2.31
- RNS: 0.90
- p -value: 0.031157
- ϕ -value: 0.2068

State results:

- FER: 60.0%
- WS: -2.10
- RNS: 0.91
- p -value: 0.014953
- ϕ -value: 0.2310

Rough interpretation. Several patterns are clearly discernible within these results, first among them the high degree of significance of the Univ and TAML features, as well as their relatively high ϕ -values (strong medium strength of association with Univ and strong strength of association with TAML¹¹), suggesting that both features are highly salient to the model's functioning. While the remaining features show solid significance by way of p -value, their ϕ -values are all relatively low (only activity approaches the threshold for medium strength of association). It is also worth noting that while Res is the best performing feature in this experiment with a WS of 22.90, its X^2 -score did not breach the threshold of significance.

MLP

The WS range for the German RFC experiment was -4.50 to 8.40 (12.90), indicating that, as with the Chinese MLP experiment, there is broad differentiation such that there are a number features which show meaningful deviation from baseline. In other words, MLP highlights multiple German features as more or less error-prone, but not with the same extreme skew as RFC.

¹¹ This TAML ϕ -value is the highest in the entire pilot experiment.

Univ results:

- FER: 73.3%
- WS: 5.04
- RNS: 0.74
- p -value: 0.00001
- ϕ -value: 0.3927

Activity results:

- FER: 71.4%
- WS: 2.26
- RNS: 0.52
- p -value: 0.009534
- ϕ -value: 0.2505

TAML results:

- FER: 62.5%
- WS: 8.40
- RNS: 1.00
- p -value: 0.00001
- ϕ -value: 0.4802

Accomp results:

- FER: 12.9%
- WS: -4.50
- RNS: 0.00
- p -value: 0.017441
- ϕ -value: 0.2108

NegPol results:

- FER: 54.5%
- WS: 2.30
- RNS: 0.53
- p -value: 0.032536
- ϕ -value: 0.2060

State results:

- FER: 60.0%
- WS: 2.46
- RNS: 0.54
- p -value: 0.015683
- ϕ -value: 0.2303

Rough interpretation. Perhaps most striking about these results is their divergence from the German RFC results, in which performance was not only at or significantly baseline but also a much wider WS range occurred. Additionally, the NegPol and activity values are nearly flipped between the two experiments. Experiment-internally, the major outliers are TAML and accomp, with their respective RNS's of 1.00 and 0.00 (indicating worst and best performance, respectively). As in the RFC experiment, both Univ and TAML receive very low p -values and relatively high ϕ -values, indicating both are highly salient to MLP as they are to RFC. The remaining features, while statistically significant, all show relatively low strength of association.

Interpretation & Discussion

There are several key observations which can be made on examination of these results: both models show a variety of significant (by statistical measures) features, most most of these (with the exception of Univ and TAML) show relatively weak strength of association; both models performed notably well with resultatives (RFC especially) though the X-square results were not deemed significant; both models struggled heavily with universals and TAM layering, though as with resultatives and experientials, the exact ratios and degrees of significance and association varied by model.

In the case of both models' overprediction of *pastperf* in the cases of *k2* and *past* forms, all instances correlates directly with the TAML feature, which in turn was shown to both be statistically significant and to bear high strength of association. Taken together, this paints a picture wherein the TAML feature causes the most difficulty for both models to predict. I propose that this owes to the coarse-grainedness of the feature itself; by definition it encompasses all cases wherein a perfect bears a secondary TAM feature (in this experiment, essentially past and modal perfects). The insight that can be gleaned from this is that the TAML is too coarse-grained and needs to be further decomposed in more nuanced categories such as "tense shifted" for past perfects and "modal" or even subcategories of modality for modal perfects.

The strong performance of the Res feature coupled with low statistical significance scores is a matter which is currently under review. Similarly, the apparent correlation patterns between Aktionsart and error type, while statistically significant, is complicated by relatively low strength of association scores. Both of these facets of the experimental results are being taken under consideration with respect to how they may inform subsequent experimental methodologies; input and feedback are very welcome!

Another pattern of note is that seen in the errors involving the Univ and activity and state features. Both models made errors in distinguishing between the present and perfect in the presence of the Univ feature which appear to correlate with activities and states. The results show that the model erroneously predicts present in the case of universal activities and perfect in the case of universal states. Between the specific patterns within this error class (i.e. a high degree of variance in which actual construction was mispredicted as either present or perfect), the low ϕ -values for both activity and state, and the relatively small sample size used here, I refrain from posing serious claims, though it seems safe to say that in German, the perfect is more closely associated with universal activities while the present is more closely associated with universal states.

The errors of type *perf* → [varies], with their more or less random feature value distributions, most likely reflect an overgeneralization on both models' part to default to the perfect in cases where specific feature profiles prove uninformative. This will be monitored on an ongoing basis as the feature schema, data sets, and overall experimental methodologies evolve.

Speaking to the methodology more broadly, two questions arise which, while perhaps straightforward to someone with more experience working with the two models, strike me as compelling: 1) What about the workings of a decision tree model (RFC) paired with the methodology employed here makes the resultatives in this data set exceptionally easy to predict? 2) Why should a neural network-based model (MLP), again paired with the methodology employed here, perform so much better on accomplishments and experientials than a decision tree model?¹²

English

The overall error rates for the two English experiments were: 7.3% of 150 tokens for RFC and 6.0% of 150 tokens for MLP.

Error Type Results

With respect to specific error types, one noteworthy pattern emerged: predicted *pastperf* → actual [varies, all modals].

***Pastperf* → [modals]**. This error type represents 75% of all RFC errors and 55.6% of all MLP errors. All instances of this error bear the TAML feature, and favor achievements at a rate of 6/8 for RFC and 4/5 for MLP.

The remaining results for both experiments showed no discernible patterning, either by error type or feature value clustering.

¹² Note to Barend: I believe I haven't actually referenced this facet of the results here yet; not to worry, I have a mountain of data and interpretations and insights to sift through, which on the one hand is extremely useful but on the other makes it easy for things to get missed. I have it all organized and will make sure everything that needs to be worked in gets worked in.

Feature-Value Results

A feature-value analysis on the results of both models revealed two features of potential significance: TAML and negative polarity (NegPol).

RFC

The WS range for the English RFC experiment was -0.70 to 5.05 (6.13), indicating that most features hover near baseline performance, with little feature differentiation.

TAML feature results:

- FER: 30.3%
- WS: 5.05
- RNS: 1.00
- p -value: 0.00001
- ϕ -value: 0.4680

NegPol feature results:

- FER: 33.3%
- WS: 2.08
- RSN: 0.52
- p -value: 0.002488
- ϕ -value: 0.2941

Rough interpretation. The results from English are fairly clean cut: in addition to the comparatively low error rate only two features show significant X-squared values, NegPol and TAML. NegPol shows a relatively solid p -value, and a ϕ -value brushing the threshold for medium-strength association (0.3), though its WS is unexceptional. TAML is by far the standout here; 9 of the 11 errors made bear this feature, and with a strong X^2 -score and ϕ -value, this feature is clearly a challenge for the model.

MLP

The WS range for the English MLP experiment was -6.36 to 0.00 (6.36), indicating that while features perform better than baseline and a low degree of feature differentiation, some features did stand out slightly.

TAML feature results:

- FER: 33.3%
- WS: -1.91
- RNS: 0.70
- p -value: 0.00001
- ϕ -value: 0.4656

NegPol feature results:

- FER: 33.3%
- WS: -0.69
- RSN: 0.70
- p -value: 0.0085
- ϕ -value: 0.2585

Rough interpretation. The results from the MLP experiment align neatly with the RFC results.

Interpretation & Discussion

Both models performed extremely well with English; not only is English the anchor language of the experiment, English is also less morphologically complex than German (though more so than Chinese), which yields a smaller set of constructions to work with, but also the vast majority of the research on PLCs has at least started with if not entirely focused on English, so the feature set in its starting state is presumably influenced by features of the English perfect specifically owing to its prevalence in the relevant literature.

Aside from this, there is a transparent correlation between errors and the TAML feature, which corroborates the insight from German that this feature deserves to be further decomposed. From a linguistic perspective, this is relatively transparent: the TAML feature encompasses both tense-shifted and modal forms (*had gone*, *might/would/could/should have gone*), and the model clearly wants to assign *pastperf* to all of them; presumably it reads *pastperf* as a “default” for TAML based on frequency (or other TBD) effects.

Additionally, NegPol in both experiments shows a relatively strong p -value, and a ϕ -value in the weak to nearly-medium strength of association range, though its WS is unexceptional. My interpretation of this facet of the results is that *something* is going on with NegPol, though with a small sample size (only 12 of 150 tokens in both experiments bear this feature) it is difficult to draw serious conclusions.

Cross-Linguistic Patterns and Insights

- 300 tokens in 3 languages is a great number for a pilot experiment, but much more data will be required for truly compelling insights; but that's okay — that's what the “real” (non-pilot) experiments are there for
- Error patterns between English and German were fairly similar, while the error patterns with Chinese were wholly distinct
 - This in some ways poses problems for the methodology — can perfects/PLCs even be considered a coherent class? — on the other hand it's extremely useful in that more (and better annotated by actual L1 speakers) data may reinforce some of the patterns that here are suggestive but not as yet truly compelling
 - This is totally expected: perfects/PLCs are notoriously “slippery” in their semantics, not only cross-linguistically but even intra-linguistically
- Unfortunately, there aren't *that* many real standouts. That said, this is a pilot experiment with a small sample size (300 tokens in three languages with a train/test split of 50/50), the features are preliminary and as such somewhat naive (they're pulled from the literature and identified by rough frequency of mention — i.e. these are some of the most commonly discussed/proposed semantic features of cross-linguistic PLCs), and were always going to need significant refinement. I just wish the signals as to *how* to refine them were a weeeeeee bit clearer...
- With respect to Chinese, both models showed a pattern of performing particularly well on resultatives and particularly poorly on experientials, even though the strength of association for both features and both models is relatively weak
 - The fact the same pattern appeared in both, however strong the association, suggests that these two features are significant for how the models interpret the data
 - From a linguistic perspective, this may mean, for example, that one particular construction is being associated (correctly in the case of resultatives, incorrectly in the case of experientials) with each of these features
 - Based on the error patterns seen in the previous section, it appears that the model defaults to *le* for resultatives, even though *suo_de* and *unmarked* also occur frequently with resultatives
 - *For analysis in the thesis: what about the Chinese TAM landscape affects this pattern?*
 - There is nothing in the error patterns that points specifically to the Exp feature; it's more or less evenly spread across error types. My read is that the Chinese TAM system is moderated by (is this the right wording? “operates on”? “is conditioned by”? “hinges on”?) a different featural “constellation” than English and German, which despite their grammatical differences, are in fact still closely related languages that have long existed in near contact with each other
 - While not picked up as significant in the X-square analysis, it is noteworthy that for both models, the *le* → *suo_de* error correlates strongly with accomplishments, and the *le* → *unmarked* error correlates just as strongly with achievements
- With respect to German, there are a few significant observations that can be made: both models show significant (by X-square analysis) features, though most of them have relatively weak ϕ -values; both models

performed notably well with resultatives (RFC especially) thought the X-square results were not deemed significant; both models struggled heavily with universals and TAM layering, though as with resultatives and experientials, the exact ratios and degrees of significance and association varied by model

- As we saw with the English results, TAM layering is a non-monolithic grouping of possibly fine-grained distinctions; the takeaway is that this feature should be decomposed going forward
- The universals are an interesting one, and if we look at the results from both models it appears that German uses both *perf* and *pres* constructions in the context of the Univ feature, and the model is struggling to identify which to use where
 - Based on the specific error patterns, it's worth pointing out that for both models, the *perf* → *pres* errors are all activities, and the *pres* → (varies) errors are all states
 - From a linguistic perspective this suggests an association in German between universals and activity/state Aktionsart values which for some reason neither model is sensitive to
- With respect to English, the results are fairly straightforward:
 - I suspect that what's happening (linguistically) with the NegPol cases is that these overlap significantly with the TAML cases, based on my (non-quantitative) impression that there are a large number of negative polarity modals (e.g. *would not have verbed*, *should not have verbed*, etc.) in the dataset, and this is skewing the NegPol results by association with the TAML feature; this is reinforced by the fact that while X-square results for NegPol are relatively significant, both the ϕ -values and WS's are fairly weak
 - As with German, TAML needs to be decomposed
- A truly global observation that needs to be made is that these data were coded by one person (me yay), and there was no shortage of edge cases, cases that were ambiguous, and best calls I could do wrt construction use/equivalence/etc. Overall, this effect shouldn't be *huge*, but at least deserves mention; there are at least two steps that I can take to address this, though I humbly invite the committee to suggest more:
 - Bring in L1 annotators: undergraduate L1 annotators are much better equipped than I am to correctly identify grammatical constructions in a given language, as well as to identify when a clause conveys equivalent (perfect-like) semantics and further when a construction conveys additional semantics that are absent in the English
 - Improve diagnostics: one of the unfortunate pitfalls of computational/corpus linguistics is that it is impossible to probe a given utterance to tease apart ambiguity or other edge cases; you just have the text on the page and the surrounding context to do your best with. My plan to address this is to continue to refine the diagnostics used in identifying a construction's feature values so as to ensure that even in instances where, say, ambiguity is unresolvable, guidelines will exist to ensure that similarly ambiguous cases will at least receive the same treatment¹³

Next Steps

Coming soon!

¹³ This is absolutely something I'm already doing, but the system can definitely stand to be improved

V. Future Work & Dissertation Plan

Core points to get across:

- ☐ (Planned) chapters (or possibly after extend modeling/before planning)
- ☐ Expand data + languages.
- ☐ Refine features (esp. TAM layering).
- ☐ Scale annotation with RAs.
- ☐ Extend modeling + interpret cross-lingual results.
- ☐ Planning
 - ☐ Potential snags + mitigation.
 - ☐ Conferences/publications.
 - ☐ Timeline (year plan).